

Enrique Martínez Martel

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Madrid, Spain



SUMMARY

Machine Learning researcher and engineer with a background in deep learning, computer vision, and generative AI. Experience designing and evaluating neural network architectures across academic research and industry.

EDUCATION

Universidad Politécnica de Madrid (UPM) <i>MSc in Data Science [Thesis Grade: 10/10]</i>	Sep. 2023 – Jun. 2025 Madrid, Spain
Escuela de Organización Industrial (EOI) <i>MSc in Big Data & Digital Business</i>	Sep. 2024 – Jun. 2025 Madrid, Spain
Universitat Politècnica de Catalunya (UPC) <i>BSc in Computer Science</i>	Sep. 2017 – Jun. 2022 Barcelona, Spain

PROJECTS

MSc Thesis: Comparative Analysis of Computer Vision Techniques	UPM - 2025
Comparative study of deep learning architectures for object detection (YOLOv10, Faster R-CNN) on a custom real-world dataset; evaluated model performance, generalization, and deployment trade-offs. Thesis grade: 10/10.	
BSc Thesis: Autonomous Driving with CARLA	UPC - 2022
Developed a perception and decision-making pipeline for autonomous driving using the CARLA simulator; covered sensor data processing, object detection, and real-time control under dynamic environments.	

EXPERIENCE

Banco Santander <i>Data Scientist - IRB Risk</i>	Jan. 2025 – Present Madrid, Spain
- Part of the IRB Risk team; involved in the generation of regulatory credit risk parameters PD (Probability of Default) and LGD (Loss Given Default) under the Basel framework. - Developed and maintained ETL pipelines and data generation workflows in Databricks (Python/PySpark); provided ad-hoc support to the LGD team including outlier investigation and preparation of responses to ECB and Banco de España inspections.	
Banco Santander <i>Data Science Talent Program - Models & Data</i>	Jun. 2024 – Dec. 2024 Madrid, Spain
- Designed and benchmarked a multimodal LLM-based classification pipeline using GPT-4/GPT-4o via Azure OpenAI; evaluated transformer architectures and prompting strategies on real operational data. - Supported the Mortgages business team through customer segmentation, EDA, and automated reporting using Python.	
Omron Industrial Automation Europe <i>Junior Software Engineer</i>	Jan. 2022 – Aug. 2023 Barcelona, Spain
Contributed to software solutions optimising industrial automation processes using C#/.NET, Visual Studio, and TFS.	

SKILLS

AI/ML : LLMs, Prompt Engineering, XGBoost, LightGBM, PyTorch, Transformers, YOLO, Scikit-learn, Streamlit
Languages : Python (PySpark, Pandas, NumPy), SQL, C#/.NET
Tools & Platforms : Apache Spark, Databricks, Azure OpenAI, Git/GitHub, Power BI, Linux, Azure DevOps
Spoken Languages : Spanish (native), English (full professional proficiency), French (basic)